

Swiggy Sales Analysis

Import Libraries

```
In [15]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
```

```
In [12]: df = pd.read_excel("C:/Users/Mohammed Riyaz/Downloads/Swiggy Sales Analysis/swiggy_
```

```
In [14]: df.head()
```

```
Out[14]:
```

	State	City	Order Date	Restaurant Name	Location	Category	Dish Name	Price (INR)
0	Karnataka	Bengaluru	2025-06-29	Anand Sweets & Savouries	Rajarajeshwari Nagar	Snack	Butter Murukku-200gm	133.9
1	Karnataka	Bengaluru	2025-04-03	Srinidhi Sagar Deluxe	Kengeri	Recommended	Badam Milk	52.0
2	Karnataka	Bengaluru	2025-01-15	Srinidhi Sagar Deluxe	Kengeri	Recommended	Chow Chow Bath	117.0
3	Karnataka	Bengaluru	2025-04-17	Srinidhi Sagar Deluxe	Kengeri	Recommended	Kesari Bath	65.0
4	Karnataka	Bengaluru	2025-03-13	Srinidhi Sagar Deluxe	Kengeri	Recommended	Mix Raitha	130.0

```
In [8]: df.tail()
```

Out[8]:

	State	City	Order Date	Restaurant Name	Location	Category	Dish Name	Price (INR)	Rating	I
197425	Sikkim	Gangtok	2025-01-25	Mama's Kitchen	Gangtok	Momos	Soya cheese chilli momo ...	112.0	4.4	
197426	Sikkim	Gangtok	2025-07-02	Mama's Kitchen	Gangtok	Momos	Kurkure momo fried ...	140.0	4.4	
197427	Sikkim	Gangtok	2025-03-25	Mama's Kitchen	Gangtok	Momos	Chilli cheese momo	126.0	4.4	
197428	Sikkim	Gangtok	2025-03-26	Mama's Kitchen	Gangtok	Momos	Veg Momos (8 Pc)	85.0	4.4	
197429	Sikkim	Gangtok	2025-03-27	Mama's Kitchen	Gangtok	Momos	Soya Momo	100.0	4.4	

Metadata

```
In [9]: print("No of Rows:", df.shape[0])
```

No of Rows: 197430

```
In [10]: print("No of Fields:", df.shape[1])
```

No of Fields: 10

```
In [11]: df.info
```

Out[11]: <bound method DataFrame.info of

	State	City	Order Date
Restaurant Name \			
0	Karnataka	Bengaluru	2025-06-29
1	Karnataka	Bengaluru	2025-04-03
2	Karnataka	Bengaluru	2025-01-15
3	Karnataka	Bengaluru	2025-04-17
4	Karnataka	Bengaluru	2025-03-13
...	...	...	...
197425	Sikkim	Gangtok	2025-01-25
197426	Sikkim	Gangtok	2025-07-02
197427	Sikkim	Gangtok	2025-03-25
197428	Sikkim	Gangtok	2025-03-26
197429	Sikkim	Gangtok	2025-03-27

	Location	Category \
0	Rajarajeshwari Nagar	Snack
1	Kengeri	Recommended
2	Kengeri	Recommended
3	Kengeri	Recommended
4	Kengeri	Recommended
...	...	...
197425	Gangtok	Momos
197426	Gangtok	Momos
197427	Gangtok	Momos
197428	Gangtok	Momos
197429	Gangtok	Momos

	Dish Name	Price (INR) \
0	Butter Murukku-200gm	133.9
1	Badam Milk	52.0
2	Chow Chow Bath	117.0
3	Kesari Bath	65.0
4	Mix Raitha	130.0
...	...	...
197425	Soya cheese chilli momo	112.0
197426	Kurkure momo fried	140.0
197427	Chilli cheese momo	126.0
197428	Veg Momos (8 Pc)	85.0
197429	Soya Momo	100.0

	Rating	Rating Count
0	4.0	0
1	4.5	25
2	4.7	48
3	4.6	65
4	4.0	0
...	...	...
197425	4.4	0
197426	4.4	0
197427	4.4	0
197428	4.4	0
197429	4.4	0

[197430 rows x 10 columns]>

Data Types

```
In [20]: df.dtypes
```

```
Out[20]: State                object
City                object
Order Date          datetime64[ns]
Restaurant Name      object
Location             object
Category             object
Dish Name            object
Price (INR)          float64
Rating              float64
Rating Count         int64
dtype: object
```

```
In [21]: df.describe()
```

```
Out[21]:
```

	Order Date	Price (INR)	Rating	Rating Count
count	197430	197430.000000	197430.000000	197430.000000
mean	2025-05-01 19:41:20.996808960	268.512920	4.341582	28.321805
min	2025-01-01 00:00:00	0.950000	1.500000	0.000000
25%	2025-03-01 00:00:00	139.000000	4.300000	0.000000
50%	2025-05-02 00:00:00	229.000000	4.400000	2.000000
75%	2025-07-01 00:00:00	329.000000	4.500000	15.000000
max	2025-08-31 00:00:00	8000.000000	5.000000	999.000000
std	NaN	219.338363	0.422585	87.542593

KPI's

Total Sales

```
In [56]: total_sales = df["Price (INR)"].sum()

print(f"Total Sales (INR): ₹{total_sales:,.2f}")
```

Total Sales (INR): ₹53,012,505.77

Average Ratings

```
In [69]: average_rating = df["Rating"].mean()
print("Average Rating:", round(average_rating,1))
```

Average Rating: 4.3

Average Order Value

```
In [73]: average_order_value = df["Price (INR)"].mean()  
print(f"Average Order value: ₹{average_order_value:,.2f}")
```

Average Order value: ₹268.51

Average Rating Count

```
In [80]: average_rating_count = df["Rating Count"].mean()  
print("Average Rating Count:", round(average_rating_count,1))
```

Average Rating Count: 28.3

Rating Count

```
In [81]: rating_count = df["Rating Count"].sum()  
print("Rating Count:", round(rating_count,1))
```

Rating Count: 5591574

Total Orders

```
In [84]: total_orders = df["Order Date"].count()  
print("Total Orders:", round(total_orders,2))
```

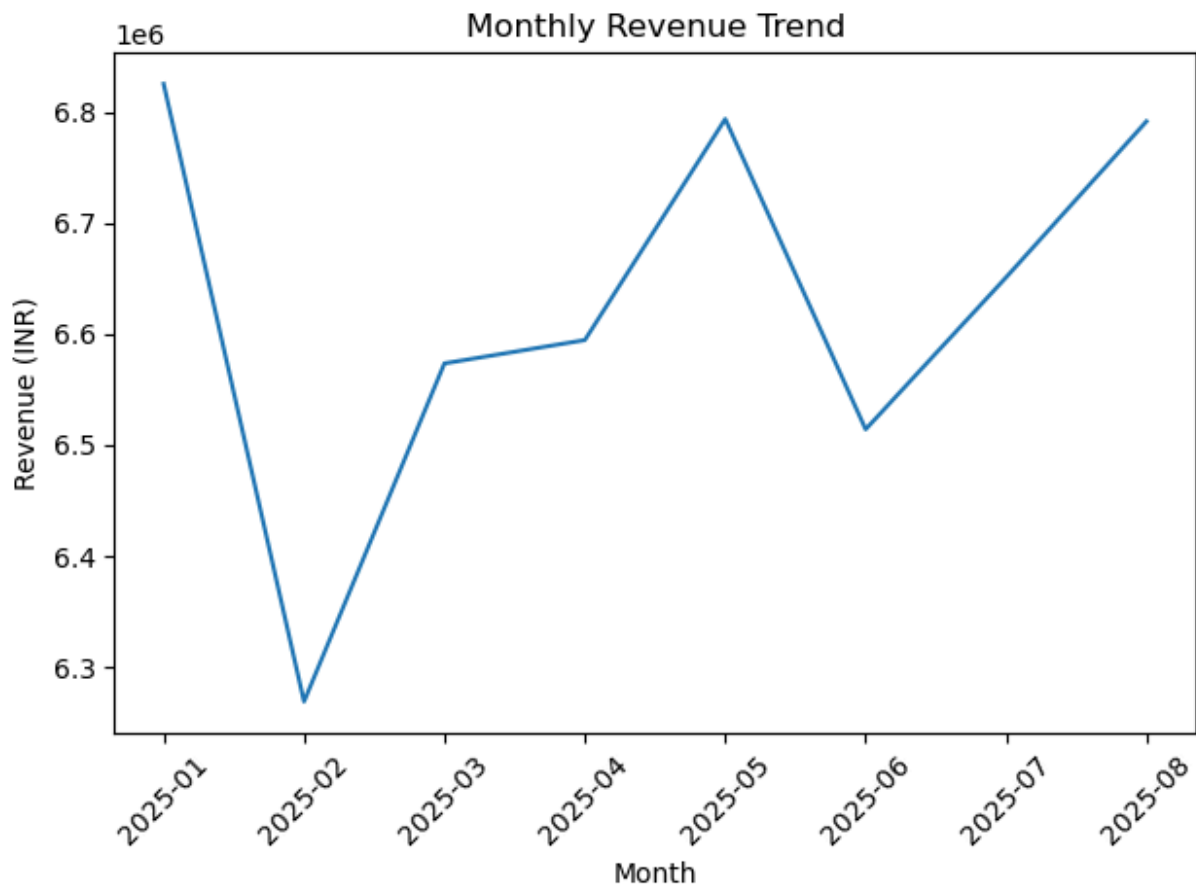
Total Orders: 197430

CHARTS Design

Montly Sales Trend

```
In [19]: df["Order Date"] = pd.to_datetime(df["Order Date"])  
df["YearMonth"] = df["Order Date"].dt.to_period("M").astype(str)  
  
monthly_revenue = df.groupby("YearMonth")["Price (INR)"].sum().reset_index()  
  
plt.figure()  
plt.plot(monthly_revenue["YearMonth"],monthly_revenue["Price (INR)"])  
plt.xticks(rotation=45)  
plt.xlabel("Month")  
plt.ylabel("Revenue (INR)")  
plt.title("Monthly Revenue Trend")  
plt.tight_layout()  
plt.show
```

Out[19]: <function matplotlib.pyplot.show(close=None, block=None)>



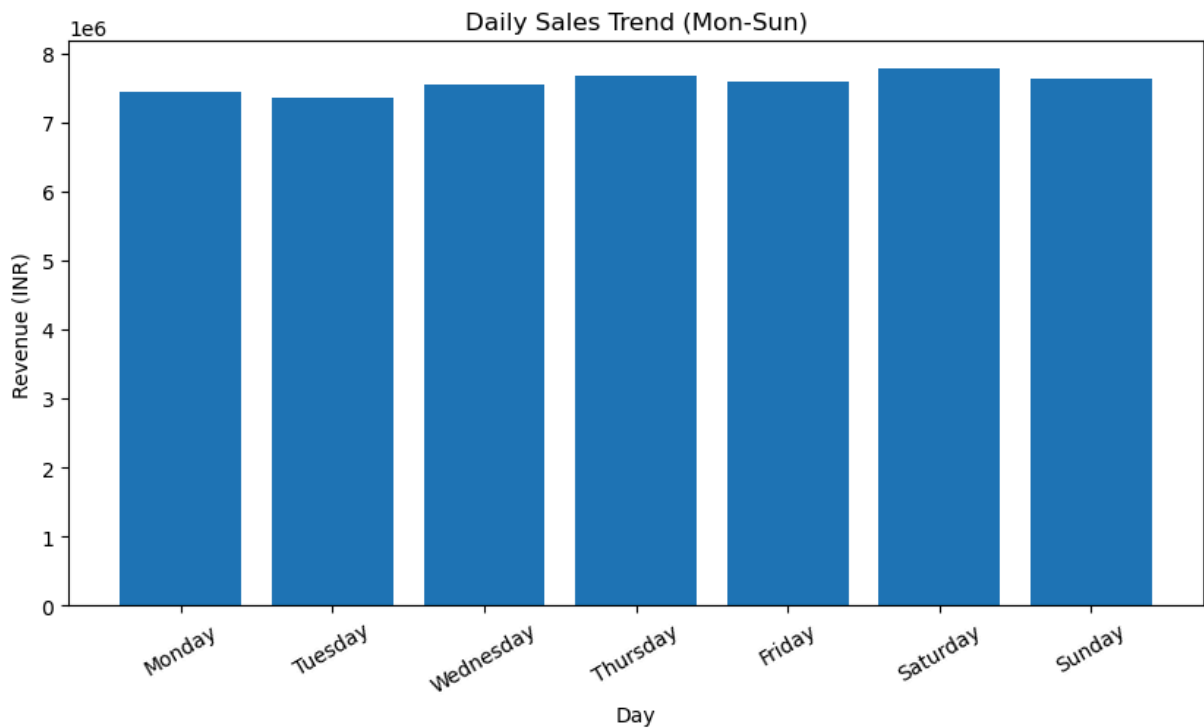
Daily Sales Trend

```
In [33]: df["DayName"] = pd.to_datetime(df["Order Date"]).dt.day_name()

daily_revenue = (
    df.groupby("DayName")["Price (INR)"]
      .sum()
      .reindex(["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"])
)

plt.figure(figsize=(10,5))
plt.bar(daily_revenue.index, daily_revenue.values)
plt.title("Daily Sales Trend (Mon-Sun)")
plt.xlabel("Day")
plt.ylabel("Revenue (INR)")
plt.xticks(rotation=30)
plt.show
```

```
Out[33]: <function matplotlib.pyplot.show(close=None, block=None)>
```



Total Sales by Food Type (Veg vs Non-Veg)

```
In [36]: non_veg_keywords = [
    "chicken", "fish", "mutton", "prawn",
    "biryani", "egg", "kabab", "kebab",
    "non-veg", "non veg"
]

df["Food Category"] = np.where(
    df["Dish Name"].str.lower().str.contains("|".join(non_veg_keywords), na=False),
    "non-veg",
    "veg"
)
```

```
In [38]: food_revenue = (
    df.groupby("Food Category")["Price (INR)"]
    .sum()
    .reset_index()
)
```

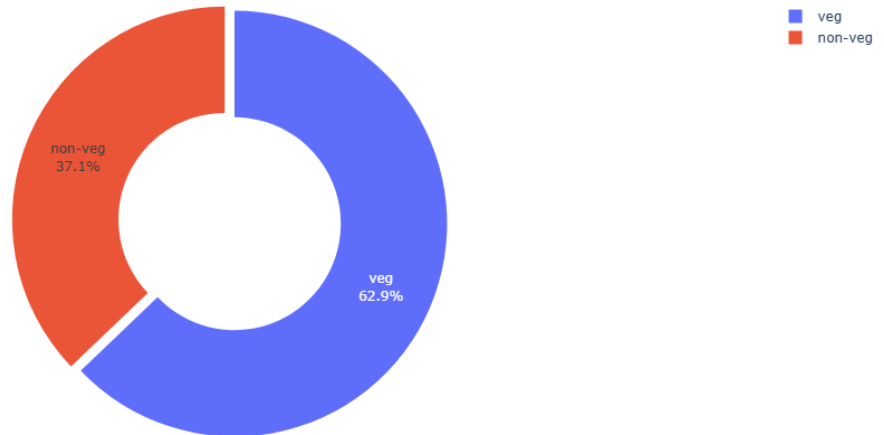
```
In [49]: fig = px.pie(
    food_revenue,
    values = ("Price (INR)"),
    names = ("Food Category"),
    hole = 0.5,
    title = "Revenue Contribution: Veg vs Non Veg",
)

fig.update_traces(
    textinfo = "percent+label",
    pull = [0.05, 0]
)
```

```
fig.update_layout(
    height=500,
    margin=dict(t=60,b=40,l=40,r=40)
)

fig.show()
```

Revenue Contribution: Veg vs Non Veg

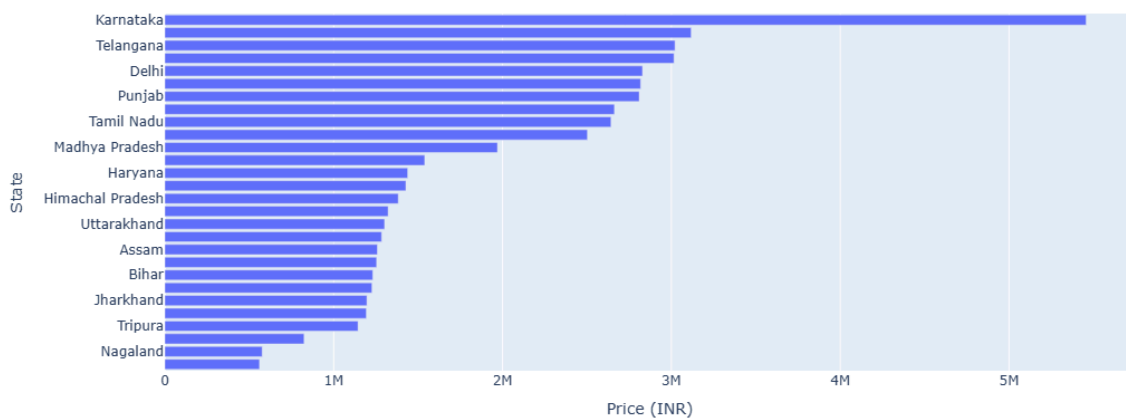


Total Sales by State

```
In [53]: fig= px.bar(
    df.groupby("State", as_index=False)["Price (INR)"].sum()
    .sort_values("Price (INR)", ascending=False),
    x="Price (INR)",
    y="State",
    orientation="h",
    title="Revenue By State (INR)"
)

fig.update_layout(height=500, yaxis=dict(autorange="reversed"))
fig.show()
```

Revenue By State (INR)



Quarterly Performance Summary

```
In [62]: df["Order_Date"] = pd.to_datetime(df["Order Date"])
df["Quarter"] = df["Order_Date"].dt.to_period("Q").astype(str)
quarterly_summary = (
    df.groupby("Quarter", as_index=False)
    .agg(
        Total_Sales=("Price (INR)", "sum"),
        Avg_Rating=("Rating", "mean"),
        Total_Orders=("Order_Date", "count")
    )
    .sort_values("Quarter")
)

quarterly_summary
```

```
Out[62]:
```

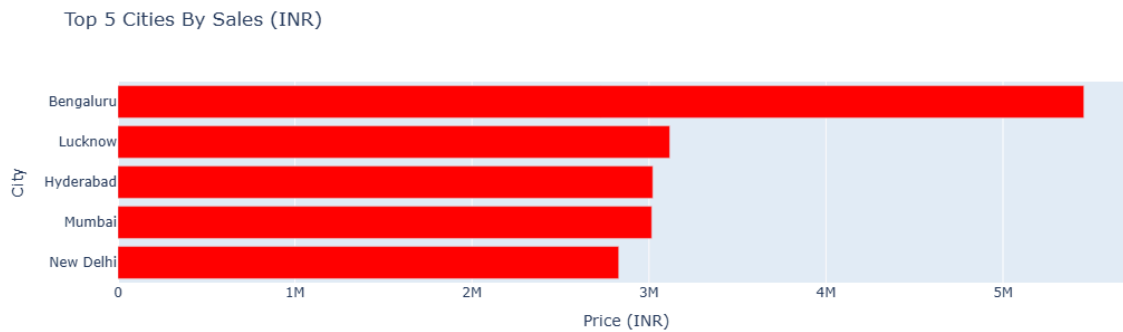
	Quarter	Total_Sales	Avg_Rating	Total_Orders
0	2025Q1	19667821.77	4.34	73096
1	2025Q2	19902256.59	4.34	74163
2	2025Q3	13442427.41	4.34	50171

Top 5 Cities by Sales

```
In [66]: top_5_cities = (
    df.groupby("City")["Price (INR)"]
    .sum()
    .nlargest(5)
    .sort_values()
    .reset_index()
)

fig = px.bar(
    top_5_cities,
    x = "Price (INR)",
    y = "City",
    orientation="h",
    title = "Top 5 Cities By Sales (INR)",
    color_discrete_sequence=["red"]
)

fig.show()
```



Weekly Trend Analysis

```
In [69]: df["Order Date"] = pd.to_datetime(df["Order Date"])
df["DayName"] = df["Order Date"].dt.day_name()

weekly_revenue = (
    df.groupby("DayName")["Price (INR)"]
      .sum()
      .reindex([
          "Monday",
          "Tuesday",
          "Wednesday",
          "Thursday",
          "Friday",
          "Saturday",
          "Sunday"
      ])
      .reset_index()
)

plt.figure(figsize=(8,5))

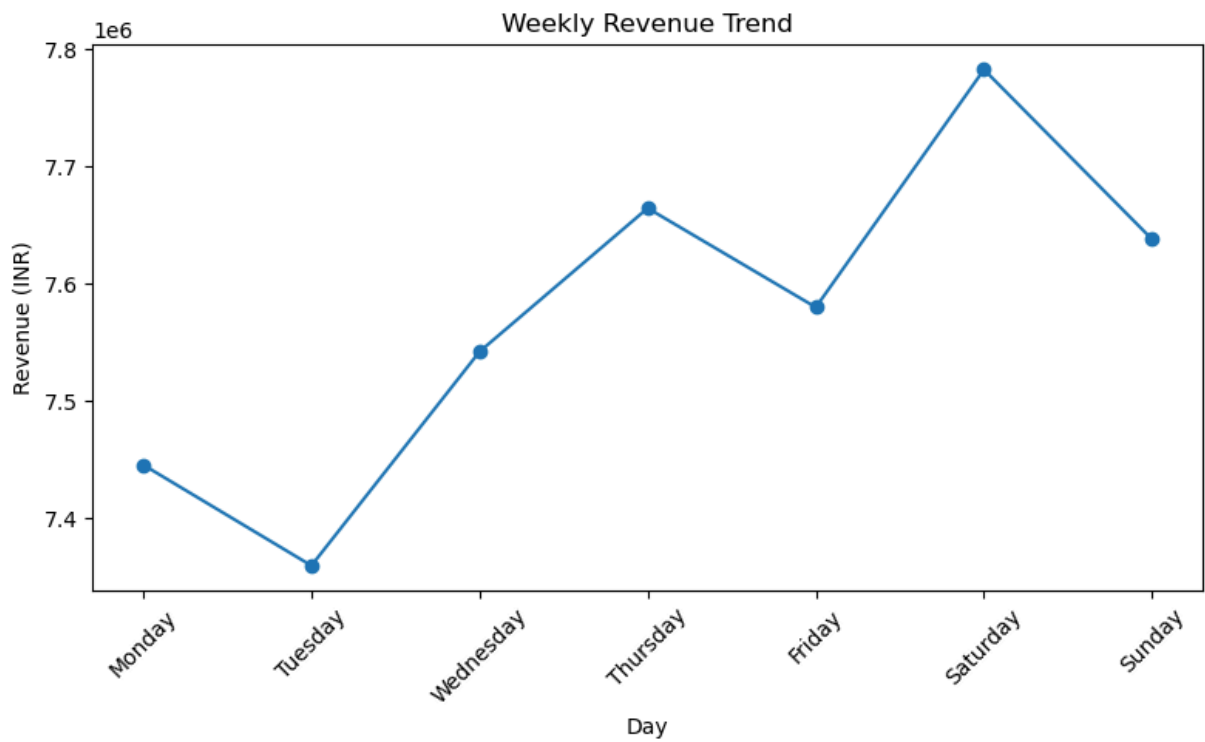
plt.plot(
    weekly_revenue["DayName"],
    weekly_revenue["Price (INR)"],
    marker="o"
)

plt.xticks(rotation=45)

plt.xlabel("Day")
plt.ylabel("Revenue (INR)")
plt.title("Weekly Revenue Trend")

plt.tight_layout()

plt.show()
```



In []: